

DDoS Anomaly Detection in Network Traffic using Quantum-Inspired Machine Learning



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Basic idea — online windowed detection

Iteratively flag DDoS using a ~1-minute window

Split each capture into tumbling windows of $W = 6\,667$ rows (≈ 1 min, 15 windows / capture).

attack bursts (ramp-up · peak · taper)



OUTPUT 1 · Window-level

Which 1-minute windows contain a DDoS burst?

Goal: alert during ramp-up, before peak.

OUTPUT 2 · IP-level

Which source IPs are responsible for the attack?

Per-IP malicious score, ranked.

100 000 flow records / 15-min capture · 2 dataset families · online streaming

Inside each window: per-IP feature engineering

33 features capturing per-IP behaviour and cross-IP coordination



zoom into one 1-min window

one instance = (src_ip, window)

Per-IP behaviour · 26 features

VOLUME	flow_count, total_packets_sum, total_bytes_sum
RATE	pps_mean / max, bps_mean / max
DURATION	duration_mean, duration_max
TIMING	inter_arrival_mean / std
SIZE	packet_size_avg_mean / std
BALANCE	outbound_ratio_mean / min
DEST.	unique_dst_ip / port, dst_concentration
PROTO.	tcp / udp / icmp share, unique_protocols
RATIOS	small_packet, high_pps, low_outbound shares

Cross-IP coordination · 7 features

the DDoS signature

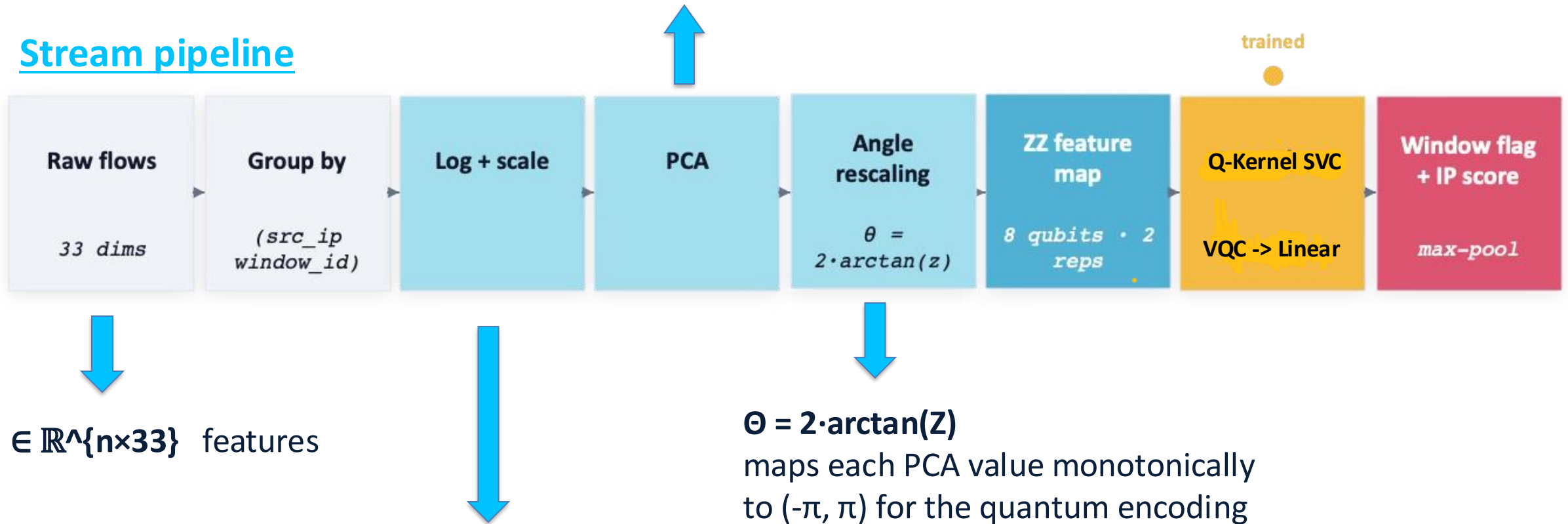
same_dst_ip_source_count
 IPs sharing the same top target
co_targeting_score
 joint flow- + source-share at top dst
window_dst_entropy
 diversity of destinations in window
+ port variants

DDoS = collective signature. A single high-rate flow is ambiguous; many sources converging on one target is not.

Architecture

PCA: **dimensionality reduction**, extracts **principal features** from network traffic data, emphasizing the most significant variations for DDoS detection.

Stream pipeline



$F_{\log} = \log(1 + F)$ on heavy-tail columns

$F_s = \text{StandardScaler.fit_transform}(F_{\log})$

Three quantum architectures

#1 — Quantum Kernel + SVC

Kernel decision rule

$$K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$$

Train:

$$G_{ij} = K(x_i, x_j) \\ \text{SVC.fit}(G, y)$$

Inference for new x^* :

$$g_i = K(x^*, x_i), i \in SV \\ f(x^*) = \sum \alpha_i y_i g_i + b$$

On quantum hardware:

each K via SWAP test or
compute-uncompute,
 $O(N^2/\epsilon^2)$ for the full Gram matrix.

#2 — Variational Quantum Circuit (VQC)

VQC layer ($k = 4$ example)

q0: $-H-$ phase(θ) $- \blacksquare -$ RY(β_0) $-$
q1: $-H-$ phase(θ) $-RZZ(\alpha_0)- \blacksquare -$ RY(β_1)
q2: $-H-$ phase(θ) $-----RZZ(\alpha_1)- \blacksquare -$ RY(β_2)
q3: $-H-$ phase(θ) $-----RZZ(\alpha_2)-$ RY(β_3)

$\uparrow$ data $\uparrow$ trainable $\uparrow$ trainable
re-upload entanglement single-qubit

Repeat L times $\rightarrow$ measure $\langle Z \rangle$ and $\langle ZZ \rangle$.

Gradients ∇ via autograd over the dense statevector simulator.

Benchmarks

Models we compare against, and metrics we report

Model	What it does	Online?
Majority class	Always predicts benign — floor baseline.	no
Classical RBF	Same windowed features → StandardScaler → RBFSampler (random Fourier feats) → balanced LogReg.	yes
PCA-Linear ablation	StandardScaler → PCA → tiny PyTorch linear scorer (skips the quantum block entirely).	yes
Bag RBF (reference)	Whole-capture per-IP aggregation → RBF LogReg. Ceiling, NOT an online detector.	no
Quantum (proposed)	ZZ-Linear / VQC / Quantum-Kernel head trained on the same instance features.	yes

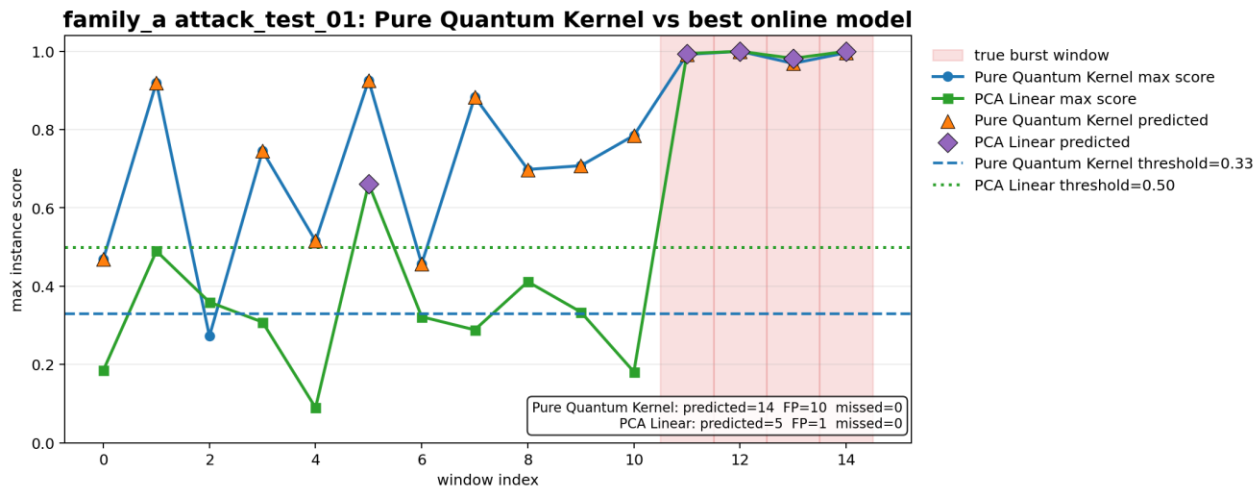
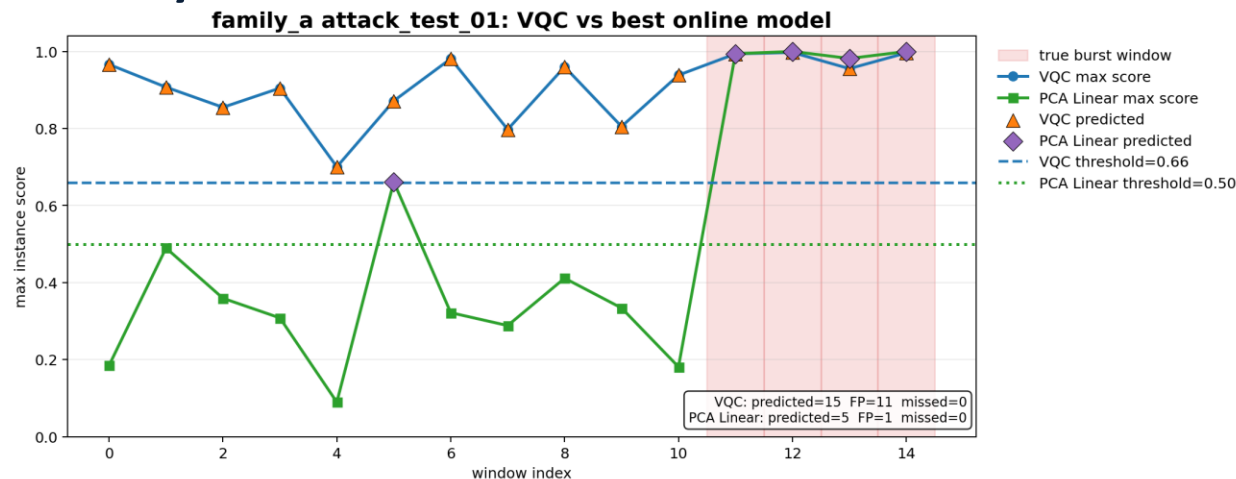
Reported metrics

- Per **evaluation level** (instance, window, dataset): **Precision, Recall, F1, AUC-ROC, full confusion matrix.**
- **Detection latency** (in windows after burst onset) — measures responsiveness.
- Calibrated **threshold τ** per family.
- Note: Bag-RBF is a CEILING reference; it is not an online window detector and should not be ranked next to streaming models.

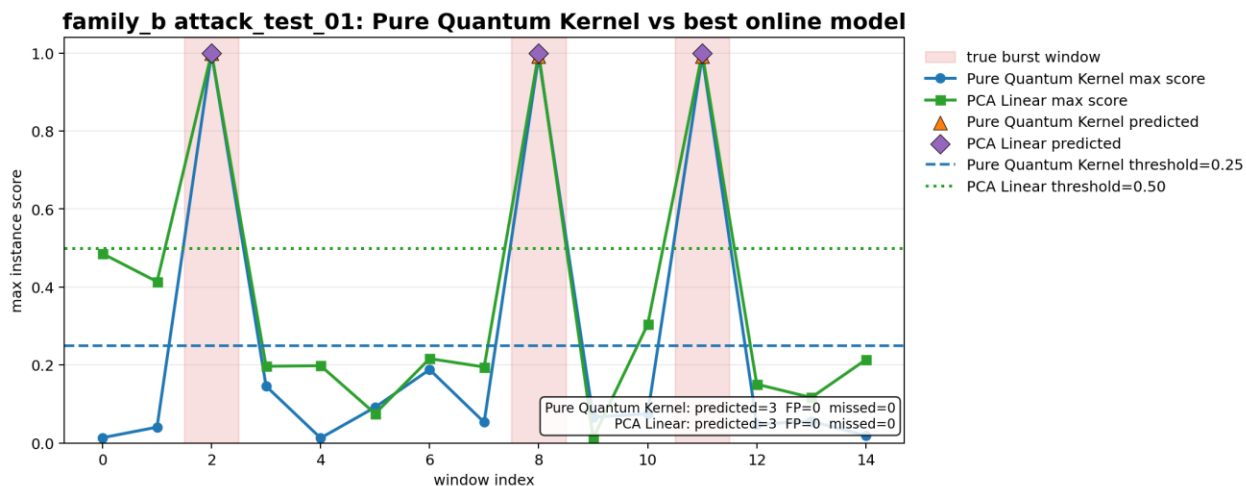
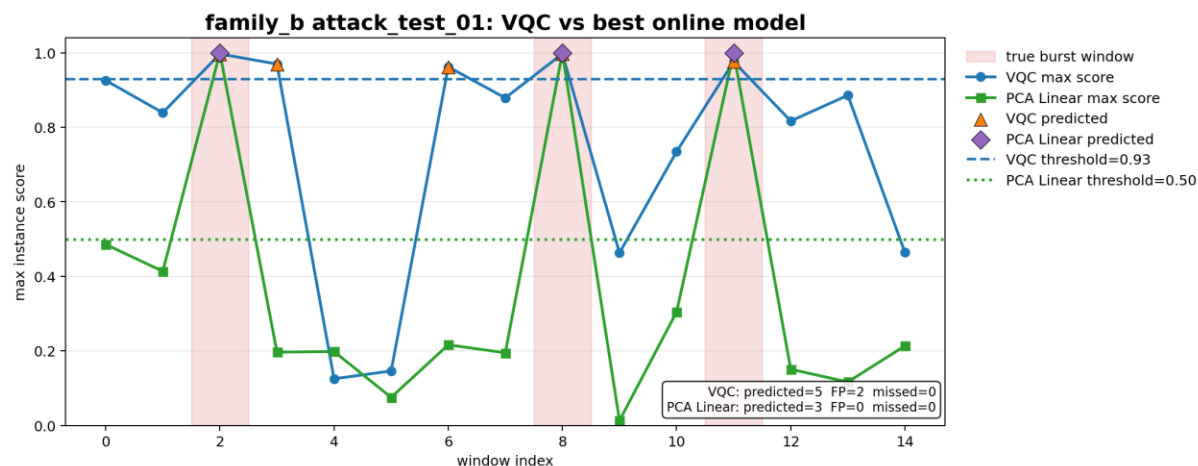
Temporal detection

Quantum (live arch) vs. best online baseline (PCA-Linear)

— Family A



— Family B



Pink shading = ground-truth burst windows. The blue line / triangles = quantum model; green line / diamonds = best online baseline. The quantum detector raises alerts inside every burst with zero latency, but spreads alerts beyond the burst region.

Quantum architecture comparison (test split)

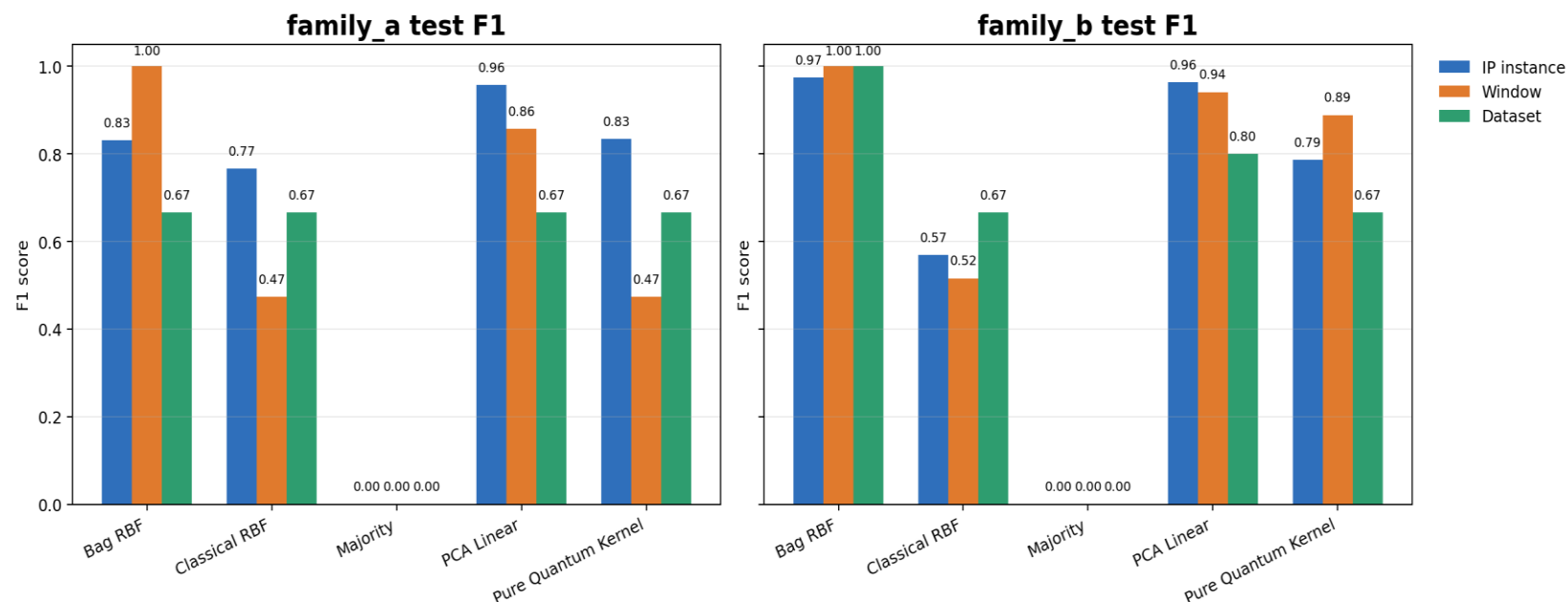
Same pipeline, three quantum blocks — picked from outputs/architectures

	VQC	Q-Kernel
Family A — threshold	0.660	0.330
Family A — Instance F1 / AUC	0.760 / 0.929	0.834 / 0.961
Family A — Window F1 / AUC	0.474 / 0.974	0.474 / 1.000
Family A — Dataset F1	0.667	0.667
Family B — threshold	0.930	0.250
Family B — Instance F1 / AUC	0.750 / 0.975	0.787 / 0.966
Family B — Window F1 / AUC	0.842 / 1.000	0.889 / 1.000
Family B — Dataset F1	0.667	0.667

Quantum-Kernel gives the best instance-level F1 in both families while VQC is the strongest compact trainable architecture

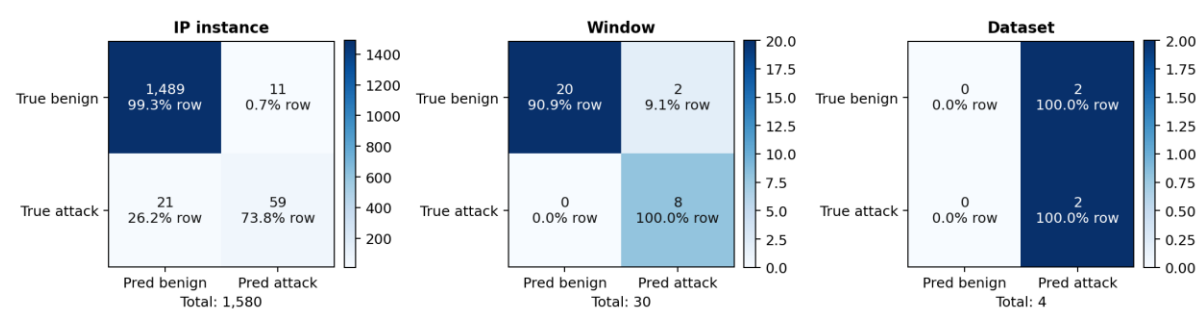
Test metrics (quantum kernel) vs. all baselines

Quantum (live arch) vs. classical baselines, all levels

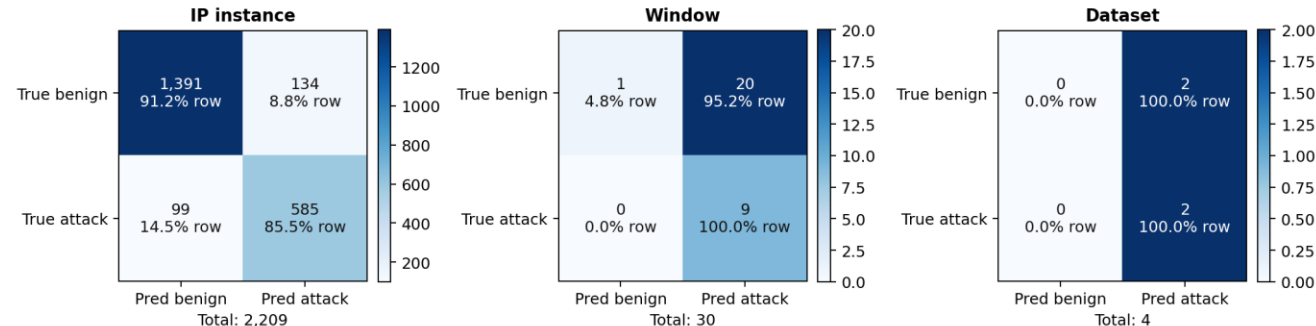


- **PCA-Linear** is the strongest among ONLINE detectors — engineered features alone are very informative.
- **QK** closes most of the gap with PCA-Linear at the instance level.
- Window F1 is dominated by precision, not recall: every model catches all bursts; the cost is **false-positive windows**.
- Bag-RBF reaches very high F1 but it is **NOT** an online detector — keep it as a ceiling.

family_b quantum pipeline test confusion matrices



family_a quantum pipeline test confusion matrices



Discussion — the surprising findings

Reading the numbers honestly

- **Surprise #1** — **PCA-Linear** (no quantum block at all) is the strongest online detector on both families.
 - Interpretation: after engineered features + StandardScaler + PCA, the residual signal is already nearly linearly separable.
 - Note: a fixed ZZ map, our first implementation cannot improve on what a single linear cut already captures.
- **Surprise #2** — quantum models do BETTER on the **non-artificial dataset** (family_b) than on the synthetic one (family_a).
 - On family_b the VQC reaches window F1 0.842 vs only 0.474 on family_a; Q-Kernel reaches 0.889 vs 0.474.
 - Hypothesis: synthetic bursts in family_a have very sharp, almost step-function feature signatures — easy for any linear model. Native DDoS in family_b has more nuanced, higher-order correlations that the entanglement layer of the ZZ map / VQC can pick up.
- **Surprise #3** — **instance AUC** is high (≈ 0.93 – 0.97 across families) but **operating-point F1** is much lower.
 - Means the score RANKING is good; the issue is the calibrated threshold + class imbalance.
- **Surprise #4** — **VQC outperforms ZZ-Linear** by a large margin in family_a (instance F1 0.760 vs 0.515).
 - Trainable RZZ + RY phases learn to undo some of the angle compression imposed by $k = 6$ PCA — confirming that fixed maps are PCA-budget bound.

Conclusions

What we can and cannot claim

- We built a **fully online** (src_ip, window) DDoS detector with three interchangeable quantum heads.
- All three architectures detect **100% of test bursts** with zero-window latency in both families.
- On family_a (synthetic) the **classical PCA-Linear ablation** is the strongest online detector — quantum heads do not add value here.
- On family_b (native DDoS) the **quantum VQC** and **Quantum-Kernel** close the gap and outperform classical RBF — first hint that **entanglement-based maps** may capture cross-IP correlations a linear classifier alone cannot.
- We do NOT claim quantum advantage:
 - Our quantum models are simulated exactly — no hardware speedup is realised.
 - PCA reduces dimensionality precisely because exact simulation requires it.
 - On FTQC we would inherit a $2^k \rightarrow \text{poly}(k)$ state-prep speedup, but **ϵ -precise expectations** still cost $\sim 1/\epsilon^2$ shots per observable. Possibility of running no dimensionality reduction.
- **Scientific takeaway:** the engineered window features carry most of the discriminative signal. A quantum feature map only pays off when its inductive bias matches the structure of the residual signal — which we do see on the native-DDoS family.

Future work

Where this proof-of-concept can grow

- **Learnable feature engineering** — replace hand-tuned thresholds (small_packet_threshold, high_pps_quantile, ...) with learnable per-window gates.
- **Joint PCA** + quantum fine-tuning — currently PCA is fit once and frozen. A simple trainable linear projector would let the quantum block reshape the angle distribution.
- **Larger k via tensor-network simulation** — go from k=6 to k=16+ classically, before relying on real FTQC hardware.
- Richer quantum feature maps — **Pauli feature maps with non-linear mixing**, problem-tailored Hamiltonian evolution embeddings.
- Per-window adaptive threshold — use validation-time recursive calibration instead of one global τ per family.
- **Stealthy attack evaluation** — low-rate, distributed, and encrypted-flow scenarios; current bursts are sharp and thus easy.
- Drift monitoring — **compute KS divergence** between cached quantum_features at train vs. live windows to detect distribution shift.

Thank you!

Questions, feedback, ideas are welcome.

<https://github.com/lollogabe/hackathon>



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